

PRESENTATION

Flutter mobile app

Citizens · Response Teams
Fire Volunteer and BFP
team captains

SOS hold · camera · GPS

Admin Console

Admin
React 18 · Vite

Accept and route
incidents · governance

Observer Console

Police · Medical
Barangay team captains
React 18 · Vite

Watch · Accept

Mapbox basemap

Raster tiles,
fetched by each
client directly

HTTPS · JSON · JWT bearer
the API is the only way in

BUSINESS

FastAPI application

REST API — 120 endpoints across 23 modules

Realtime channel (WebSocket)

Clustering

Verification

Notification

Incident
lifecycle

Summarisation

Audit

JWT validation · authority enforcement (user_role × agency_type) · request routing

in-process calls, Pydantic-validated models

DATA

Data access layer

asyncpg
connection pool

Pydantic v2
validation

Centroid ·
haversine

PostGIS proximity
(300 m radius)

Overlap
detection

Confidence
 $0.4N + 0.3S + 0.3V$

asyncpg over TLS · service role

server-side HTTPS

INFRASTRUCTURE

Supabase (managed)

PostgreSQL 17.6
+ PostGIS 3.3.7

GoTrue
authentication

Storage —
private buckets

Realtime
channels

Row-level security on all 30 tables; authority reinforced by database triggers

External services

Firebase FCM — push

Twilio — SMS OTP

Brevo — email

Anthropic — summaries

Didit — KYC review

Solid arrows are calls inside the system; dashed arrows and dashed borders are direct HTTPS to a third party.

Each layer depends only on the one beneath it. The FastAPI application holds the sole database credential, so no client reaches Supabase directly.